

```
In [4]: %runfile /Users/saminbahizad/Downloads/hmeq_eda_outputs/Assignment_4.py --
wdir

FLAG DATA
TRAINING = (4768, 30)
TEST      = (1192, 30)

AMOUNT DATA (Defaults w/ Loss Only)
TRAINING = (941, 30)
TEST      = (248, 30)
TREE CLASSIFICATION ACCURACY
=====
TREE_Train = 0.8928271812080537
TREE       = 0.8875838926174496
-----

TREE RMSE ACCURACY
=====
TREE_Train = 4561.00912099753
TREE       = 5783.259789318173
-----

RF CLASSIFICATION ACCURACY
=====
RF_Train = 0.9987416107382551
RF       = 0.9093959731543624
-----

RF RMSE ACCURACY
=====
RF_Train = 1243.4717262138286
RF       = 3280.7629671001787
-----

GB CLASSIFICATION ACCURACY
=====
GB_Train = 0.9234479865771812
GB       = 0.9043624161073825
-----

GB RMSE ACCURACY
=====
GB_Train = 1190.3722238493135
GB       = 2619.6672860537974
-----

REG_ALL CLASSIFICATION ACCURACY
=====
REG_ALL_Train = 0.8936661073825504
REG_ALL       = 0.8901006711409396
```

REG_ALL RMSE ACCURACY

=====

REG_ALL_Train = 3613.4726552849975

REG_ALL = 3493.1383897015803

TensorFlow version: 2.21.0

Training TF_relu_1HL ...

TF_relu_1HL CLASSIFICATION ACCURACY

=====

TF_relu_1HL_Train = 0.8674496644295302

TF_relu_1HL = 0.87248322147651

Training TF_relu_1HL_LOSS ...

TF_relu_1HL_LOSS RMSE ACCURACY

=====

TF_relu_1HL_LOSS_Train = 5454.503643779148

TF_relu_1HL_LOSS = 7580.2941895417225

Training TF_tanh_2HL ...

TF_tanh_2HL CLASSIFICATION ACCURACY

=====

TF_tanh_2HL_Train = 0.8026426174496645

TF_tanh_2HL = 0.790268456375839

Training TF_tanh_2HL_LOSS ...

TF_tanh_2HL_LOSS RMSE ACCURACY

=====

TF_tanh_2HL_LOSS_Train = 17136.84871847797

TF_tanh_2HL_LOSS = 17638.311937370876

Training TF_sigmoid_1HL ...

TF_sigmoid_1HL CLASSIFICATION ACCURACY

=====

TF_sigmoid_1HL_Train = 0.19714765100671142

TF_sigmoid_1HL = 0.2080536912751678

Training TF_sigmoid_1HL_LOSS ...

TF_sigmoid_1HL_LOSS RMSE ACCURACY

=====

TF_sigmoid_1HL_LOSS_Train = 17137.918660094056

TF_sigmoid_1HL_LOSS = 17639.395000963043

```
-----  
  
Training TF_relu_2HL_Drop20 ...  
TF_relu_2HL_Drop20 CLASSIFICATION ACCURACY  
=====
```

Model	TF_relu_2HL_Drop20 Train	TF_relu_2HL_Drop20
	0.7888003355704698	0.7776845637583892

```
-----
```

```
Training TF_relu_2HL_Drop20_LOSS ...  
TF_relu_2HL_Drop20_LOSS RMSE ACCURACY  
=====
```

Model	TF_relu_2HL_Drop20_LOSS Train	TF_relu_2HL_Drop20_LOSS
	17137.26232512066	17638.74734781356

```
-----
```

```
BEST TF DEFAULT MODEL: TF_relu_1HL AUC = 0.863  
BEST TF LOSS MODEL: TF_relu_1HL_LOSS RMSE = 7580.29  
ALL MODELS - LOSS AMOUNT RMSE (Test) - Tree , Regression andTensorFlow  
=====
```

Model	GB	RF	REG_ALL	TREE	TF_relu_1HL_LOSS
	2619.6672860537974	3280.7629671001787	3493.1383897015803	5783.259789318173	7580.2941895417225

```
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```

Done. All charts saved.

In [5]: